

Robert Guerra

Senior Data Engineer

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Summary

Senior Data Engineer with **13+ years of experience** designing, building, and operating **high-scale, cloud-native data platforms** supporting real-time decisioning, analytics, and **AI workloads**. Deep expertise in **Databricks Lakehouse, Apache Spark, Kafka, Airflow, Snowflake, SQL, and Python**, across **Azure, AWS, and GCP**. Proven track record delivering **low-latency streaming pipelines**, governed analytical datasets, and secure, production-grade data systems. Strong collaborator who partners with product, **ML**, analytics, and platform teams to deliver reliable, cost-efficient, and compliant data solutions in fast-paced enterprise environments.

Skills

- **Data Engineering:** ETL/ELT pipelines, batch & real-time processing, Spark, Databricks Lakehouse, Delta Lake, Kafka, Airflow
- **Cloud & Storage:** Azure, AWS, GCP, S3, ADLS, Redshift, BigQuery
- **Programming & Querying:** Python, Java, SQL, API integration (REST/gRPC), data modeling
- **Analytics & BI:** dbt, Tableau, Superset, KPI design, ML/AI dataset support
- **AI/ML Enablement:** MLflow, RAG pipelines, feature engineering, LLM telemetry support
- **DevOps & Observability:** CI/CD, monitoring, alerting, SLA management, performance tuning
- **Security & Governance:** Data governance, RBAC/ABAC, encryption, PII masking, compliance
- **Collaboration & Leadership:** Cross-functional teamwork, mentoring, agile/Scrum, documentation,

Experience

Senior Data Engineer

Infosys / Austin, TX

Nov 2022 – Present

- Architected and operated **enterprise-scale batch** and **real-time data platforms** using **Apache Spark, Databricks, Kafka**, and cloud-native storage to support operational intelligence and **AI-driven analytics**.
- Designed and implemented **Databricks Lakehouse architectures** leveraging **Delta Lake, Unity Catalog**, schema evolution, and optimized partitioning to deliver reliable, low-latency analytics.
- Built **Spark Structured Streaming pipelines** consuming high-throughput Kafka topics to power near real-time dashboards and event-driven decision systems.
- Established **Airflow-based orchestration frameworks** with **SLA enforcement**, dependency management, retries, backfills, and alerting for mission-critical data workflows.
- Led **multi-cloud data initiatives** across **Azure** and **AWS**, including migrations, hybrid ingestion strategies, and cross-cloud data access patterns.
- Developed robust **ETL and ELT pipelines in Python and advanced SQL**, ingesting data from APIs, relational databases, files, and streaming sources.
- Modeled and curated **analytics-ready datasets** to support **BI reporting, product features, operational KPIs**, and **downstream ML and AI workloads**.

- Partnered with application engineering teams to expose data via **REST** and **gRPC interfaces**, using schema-driven contracts.
- Implemented **data quality frameworks**, observability metrics, and anomaly detection to proactively identify pipeline failures and data integrity issues.
- Enforced **enterprise security and compliance controls**, including **RBAC**, least-privilege access, encryption in transit and at rest, and PII masking/tokenization.
- Built and maintained **CI/CD pipelines** using **GitHub Actions** and **Azure DevOps** to automate testing, validation, and deployment of data infrastructure.
- Tuned **Spark jobs**, **SQL** queries, and storage layouts to improve performance, reduce compute costs, and meet strict latency and throughput targets.
- Supported **AI and advanced analytics teams** by delivering feature-ready datasets, experiment metrics, and production data feeds.
- Authored detailed **architecture documentation, data lineage, and operational runbooks** to improve platform reliability and audit readiness.
- Served as a technical leader and mentor, conducting design reviews, code reviews, and driving best practices across cross-functional data teams.

Data Engineer

Dropbox / San Francisco, CA

Feb 2018 – Oct 2022

- Designed and maintained **scalable data pipelines** using **Spark** and **Kafka** to process high-volume product and event data.
- Built and optimized **Snowflake** and **BigQuery data models** to support large-scale analytical and reporting workloads.
- Developed **dbt-based transformation layers** with advanced **SQL**, including window **functions**, **CTEs**, and **performance-optimized joins**.
- Integrated data from distributed internal services and third-party APIs, ensuring consistency and correctness across systems.
- Partnered closely with product managers and analysts to translate business questions into reliable, analytics-ready datasets.
- Implemented **monitoring, logging, and alerting** to ensure high availability and rapid incident response for critical pipelines.
- Optimized storage and compute usage through partitioning, clustering, and lifecycle management strategies.
- Supported experimentation frameworks by delivering clean, well-documented datasets for feature analysis and A/B testing.
- Collaborated in a highly distributed, agile engineering environment with a strong culture of peer review and ownership.
- Contributed to internal data platform tooling to improve developer productivity and operational reliability.

Junior Data Engineer

Netflix / Los Gatos, CA

Nov 2016 – Jan 2018

- Built and maintained large-scale data pipelines processing high-volume user interaction and event data.
- Worked with distributed processing frameworks to support batch analytics and exploratory data analysis.
- Developed complex SQL transformations and optimized queries for performance at scale.
- Partnered with data scientists to deliver datasets supporting experimentation and personalization initiatives.
- Implemented validation checks and reconciliation logic to ensure analytical accuracy.
- Collaborated with engineering and analytics teams in a fast-paced, product-centric environment.
- Gained foundational experience designing data systems for globally scaled consumer platforms.

Education

The University of Chicago

Bachelor's degree in computer science

Chicago, IL

2012 – 2016